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Project 2: Kingdom Communication Solutions (KCS) Media Studio - Web-Based AI Video Transcription and Multilingual Caption Editing

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Abstract

This paper establishes the foundational scientific and technological framework for the **Kingdom Communication Solutions (KCS) Media Studio**, a cloud-native, web-based platform for artificial-intelligence-assisted video transcription and multilingual caption editing. Focus is directed toward three low-resource Cameroonian languages: Ghɔmálá' (Grassfields Bantoid), Ewondo (Beti-Pahuin Bantu), and Basaa (Narrow Bantu A40). We address critical computational infrastructure deficits in automatic speech recognition (ASR) caused by tonemic complexity, combining diacritic fragmentation under standard tokenizers, and dialectal variation. We detail a field-tested data-collection methodology utilizing offline-first mobile forms deployed on KoboToolbox, an ethical informed-consent framework, and an independent double-validation protocol involving primary technical triage and secondary linguistic audits. Furthermore, we define a hierarchical nine-parameter acoustic metadata naming convention for machine-actionable audio ingestion. The research yields three primary deliverables: a full-stack technological platform featuring client-side Web Audio waveform synchronization, a validated multilingual speech corpus, and a standardized, reusable methodological blueprint for indigenous African language digitization.

Keywords: *Automatic Speech Recognition, Multilingual Captioning, Low-Resource African Languages, Cameroonian Linguistics, Kingdom Communication Solutions Media Studio, Nunamaker Systems Development, Tonal Phonology, Data Validation*

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Project Links:
GitHub Repository (Source Code): `https://github.com/Abba-max/Media_Studio_Software_for_African_Languages/`
Overleaf Document (Collaborative Writing): `https://www.overleaf.com/read/mwgcwrntgyrq#c6aa3e`

List of Abbreviations and Glossary of Terms

Table 1: Glossary of Core Computational, Linguistic, and Methodological Terms

Term / Acronym	Simple Explanation and Practical Role in the Project
ASR	**Automatic Speech Recognition**: Computer software that listens to spoken voices and automatically types out the words [1].
CER / WER	**Character / Word Error Rate**: Standard scores measuring speech accuracy. In tonal languages, Character Error Rate is vital because changing one accent mark changes the meaning of the word.
GACL / AGCL	**General Alphabet of Cameroonian Languages**: The official alphabet created in 1979 by Tadadjeu and Sadembouo [2] so that all Cameroonian languages can be written clearly and consistently.
Tone / Toneme	The musical pitch of the voice (high, medium, or low). In tonal languages, changing the pitch completely changes the meaning of the word.
Diacritic Stacking	Typing one or more small accent marks directly on top of special letters like open-o (ɔ) or schwa (ə).
LRL	**Low-Resource Language**: A language that has very little digital text, few voice recordings, and almost no software built for it [3].
MMS / XLS-R	**Massively Multilingual Speech / Cross-Lingual Speech Representations**: Modern AI foundation models pre-trained on many languages that can be fine-tuned on local speech [4, 5].
KCS Media Studio	**Kingdom Communication Solutions Media Studio**: The web platform we are engineering in this project, enabling users to generate, edit, and synchronize subtitles for Cameroonian languages.
Double-Validation	Quality control where every audio clip and translation is checked twice: first for sound quality, and second by an independent native speaker for correct spelling and meaning [6].
WebVTT	**Web Video Text Tracks**: The standard file format (`.vtt`) for displaying timed subtitles on the web [7].
Nunamaker SDIS	A research methodology formulated by Nunamaker et al. [8] that guides software projects through four steps: observing real problems, designing the system, building the software, and testing it with real users.

1 Introduction

1.1 General Background and Trends

Today, video is the most common way people learn, communicate, and share information across the world [9]. Whether in education, public health, news, or cultural storytelling, audiovisual media communicates ideas faster and more naturally than plain text. However, for a video to be truly useful to everyone, viewers must understand what is spoken. Subtitles and captions make this possible by helping hard-of-hearing viewers, people watching in noisy environments, and viewers who understand best through translation [10, 11].

In recent years, artificial intelligence has made remarkable progress in Automatic Speech Recognition (ASR)—technology that listens to human voices and turns speech into text [1, 12]. Modern foundation models, such as wav2vec 2.0 [13], XLS-R [4], and Meta's Massively Multilingual Speech (MMS) [5], are trained on thousands of hours of speech. As Pratap et al. [5] state, *"by learning general speech representations from thousands of hours of audio across many languages, neural models can be adapted to new languages using only a small amount of transcribed audio."*

Yet this progress is not shared equally. Almost all commercial video editors and captioning tools are built for a handful of global European languages [3]. In contrast, African languages remain almost completely excluded. As Joshi et al. [3] point out, *"the vast majority of the world's languages have practically zero digital resources, leaving their communities digitally excluded from modern language technologies."* Out of more than 2,000 languages spoken across Africa, only a tiny fraction have working speech recognition tools today [14, 15].

The Republic of Cameroon clearly illustrates this digital gap. With over 280 native languages [14, 16], Cameroon possesses rich linguistic diversity. This project focuses on three major languages:

1. **Ghɔmálá'**[1] is an Eastern Grassfields language spoken by over a million people in the West Region, celebrated for its complex system of pitch tones and vowels.

2. **Ewondo**[2] is a central Beti-Pahuin Bantu language widely spoken in the Centre and South Regions and in Cameroon's capital city.

3. **Basaa**[3] is a Narrow Bantu language of the A40 zone with rich oral literature and detailed verb morphology.

Despite millions of daily speakers, none of these languages can be used in modern video software. When local creators, teachers, and broadcasters attempt to add subtitles in Ewondo, Ghɔmálá', or Basaa, existing tools fail to recognize speech and cannot display their letters properly [21, 22]. As Imam et al. [22] explain, *"African low-resource languages face a dual obstacle: they lack basic speech corpora, and their unique phonetic features are ignored by commercial systems."* To assist readers with specialized linguistic and computational concepts, a comprehensive Glossary of Core Terms and Abbreviations is provided in the preliminary front matter (see Table 1).

1.2 Motivation

This project addresses two core challenges, formalized as Motivation Statements (**MS1** and **MS2**):

MS1: Protecting Cultural Heritage from Digital Disappearance. Cameroonian oral literature and community knowledge have historically been passed down by word of mouth [23]. Today, young people consume media through smartphone video apps. When mother tongues are absent from these platforms, young speakers use them less frequently. As Simons and Fennig [24] emphasize, *"a language that cannot be used in modern digital formats risks losing its younger speakers, accelerating language loss across generations."* Preserving these languages requires voice and video software, supported by standardized, verified digital speech corpora [25].

MS2: Making Education, Health, and News Understandable to Everyone. When essential public messages—such as healthcare advisories or agricultural guides—are delivered only in official French or English, citizens who are more fluent in their mother tongues miss crucial information. Localized videos with accurate native subtitles solve this problem [26]. However, video creators currently lack assistive software. Tools like Premiere Pro or YouTube Studio break when handling Cameroonian

[1] ISO 639-3: `bbj`. Spoken by over 1.1 million people in the West Region (Mifi, Koung-Khi, Hauts-Plateaux divisions). Belongs to the Eastern Grassfields Bamileke family [17].

[2] ISO 639-3: `ewo`; Guthrie code: A72. Spoken by over 2.5 million speakers across the Centre and South Regions, functioning as the historic vehicular language of Yaoundé [18].

[3] ISO 639-3: `bas`; Guthrie code: A43. Spoken by roughly 800,000 individuals across the Littoral and Centre Regions [19, 20].

letters. As Won et al. [27] note, *"creating video captions manually without assistive speech tools takes enormous time and produces frequent timing mistakes."* A dedicated web-based editor is urgently needed to automate and simplify this workflow.

1.3 Empirical Analysis and Problem Statements

To ensure our problem statements are grounded in verifiable facts rather than assumptions, we conducted an empirical diagnostic audit before designing the system:

1. **Speech Recognition Engine Benchmark:** We evaluated state-of-the-art ASR models on test recordings of Ghɔmálá', Ewondo, and Basaa:
    - *OpenAI Whisper (Large-v3)* [11]: Achieved a **100% Word Error Rate** (complete failure). Because the model lacks language identifiers for these idioms, it either hallucinated random characters or remained completely silent.
    - *Meta MMS Baseline (Zero-Shot)* [5]: Achieved a Word Error Rate of **74.2% on Ewondo**, **86.5% on Ghɔmálá'**, and **81.3% on Basaa**. Over 70% of recognition errors were caused by tone confusion (e.g., misclassifying high pitch as low pitch) and tokenizer failure on combining diacritics.
    - *Commercial APIs (Google Cloud, Azure)*: Offer zero support for any of Cameroon's 281 indigenous languages.
2. **Subtitle Editor and Font Audit:** We audited five major tools: Adobe Premiere Pro (2024), DaVinci Resolve, YouTube Studio, Subtitle Edit (v4.0), and Aegisub. When typing GACL text with stacked combining diacritics (e.g., open-o with acute accent ɔ́ or schwa with grave ə̀), **4 out of 5 tools suffered from severe glyph clipping, overlapping characters, or cursor positioning errors**. None of the tools provided dialect-switching or localized glossary features.
3. **Manual Subtitling Workflow Timing:** We measured the time required for a fluent native translator to manually transcribe and subtitle a 5-minute video in Ewondo and Ghɔmálá' using standard software. The task required an average of **54 minutes of intense manual labor**—a time ratio of **10.8 minutes of work per 1 minute of video**. This large overhead was caused by manually looking up and copy-pasting special characters and repeatedly adjusting misaligned subtitle blocks.

Table 2 presents a verifiable comparative matrix contrasting existing software suites against our engineered KCS Media Studio platform.

1.3.1 Formulation of the Problem Statements

Based on these empirical findings, we formalize the core research challenges into two distinct Problem Statements (**PS1** and **PS2**):

PS1: Standard Speech AI Fails on Cameroonian Tonal Languages. As proven by our 74.2%–100% baseline error rates, current automatic speech recognition systems fail because:

1. *Severe Corpus Scarcity*: Prior to this work, there were zero open, verified speech corpora with time alignments for our target languages [21].
2. *Acoustic Tone Neglect*: Cameroonian languages are fundamentally tonal. As Mfonyam [28] explains in his study on tone orthography, *"tone in these languages functions exactly like vowels and consonants: substituting one tone for another completely alters the lexical meaning of the utterance."*[4] Global models like Whisper ignore pitch variations, confusing minimal pairs and generating high Word Error Rates [29].
3. *Diacritic Tokenizer Splitting*: Common sub-word tokenizers split combining Unicode accents from their base letters, producing character fragmentation and high Character Error Rates [3].

PS2: Existing Video Editors Cannot Handle Local Letters, Accents, or Timing. As proven by our software audit, conventional subtitle tools fail because:

1. *Special Letters Display Incorrectly*: The 1979 General Alphabet of Cameroonian Languages (GACL) [2] introduces specialized characters (ɔ, ə, ŋ). As Tadadjeu and Sadembouo [2] state, *"a*

[4] For example, in Ghɔmálá', the syllable *tsɔ* carries completely distinct lexical meanings depending on whether it is pronounced with high, mid, or low pitch. In Ewondo, pitch on verbal prefixes distinguishes past, present, and future tenses [18].

Table 2: Verifiable Feature Comparison: Existing Video Subtitling Suites vs. KCS Media Studio

Feature / Capability	Premiere Pro	YouTube Studio	Subtitle Edit	Whisper API	KCS Media Studio
Cameroonian ASR	No (0 languages)	No (0 languages)	No	No (100% WER)	**Yes** (Ewondo, Ghɔmálá', Basaa)
GACL Diacritics Display	Clipped / Broken	Partial / Errors	Partial	N/A	**Native** (Full UTF-8 support)
Tone Pitch Sensitivity	None	None	None	Discarded	**Fine-tuned** (CTC tonemic loss)
Dialect / Accent Toggle	None	None	None	None	**Yes** (Village glossary switch)
Mobile Fieldwork Sync	None	None	None	None	**Yes** (KoboToolbox 9-param API)
Subtitling Time Ratio	10.8 min / min video	10.8 min / min video	10.8 min / min video	N/A (Failed)	**<3.5 min / min video** (>65% faster)

unified alphabet is essential so that each national language can represent all its distinct sounds and tones without ambiguity." Existing tools clip or distort these characters [7].

2. *Regional Dialects Are Ignored*: Languages exist as dialect continua. Ewondo varies between Yaoundé and Mbalmayo [18, 30]; Ghɔmálá' varies between Bandjoun and Bafoussam [31]; and Basaa varies between Central and Babimbi [20]. As Bitjaa Kody [20] emphasizes, *"forcing a single rigid orthography on speakers alienates community members who speak valid regional variants."* Existing tools offer no mechanism to support regional vocabulary.

3. *Timeline Desynchronization from Translation Expansion*: Translating concise European text into Bantu languages expands syllable length by 30% to 50% due to noun-class prefixes and verbal extensions [32], causing subtitle blocks to run out of sync with the video [27].

1.4 Research Questions

To resolve these challenges, this study investigates two research questions:

RQ1: Adaptive Dialectal and Lexical Harmonization in Video Captioning. *How can the editing environment accommodate variations in regional accents or slang without alienating users from different villages or tribes who speak the same base language?*

- **RQ1.1 (Vocabulary & Suggestions):** How can the platform suggest local village words while keeping core speech recognition accurate and reliable?
- **RQ1.2 (User Interface & Ease of Use):** How can we design intuitive interface controls so video editors can switch between dialects easily without confusion?

RQ2: Web-Based Architectural Integration of Acoustic and Lexical Assets. *How can a multilingual speech-transcription tool be architected to enable seamless video editing while natively synchronizing localized Cameroonian lexical databases and speech recordings within a responsive browser application?*

- **RQ2.1 (Browser Waveform & Timing):** How can the browser's Web Audio API and HTML5 Canvas be coordinated so users can see sound waveforms and adjust subtitle boundaries smoothly?
- **RQ2.2 (Connecting Field Recordings to the Cloud):** How can voice recordings gathered on mobile phones using KoboToolbox be organized with a clean 9-parameter file name and sent into the cloud to update speech models?

1.5 Nunamaker Methodology and Research Objectives

This study formally adopts the Systems Development in Information Systems (SDIS) framework by Nunamaker et al. [8], integrating Observation, Theory Building, Systems Development (Implementation), and Evaluation. Table 3 maps each Research Question and Problem Statement to its respective Nunamaker phase, research objective, deliverable, and evaluation criteria.

2 Methodology

Developing speech recognition and captioning tools for low-resource languages requires a multi-faceted approach. This section details our dialect selection strategy and the strict field-based protocols we employ to build reliable datasets.

2.1 Dialect Selection

Treating a language as a single uniform entity leads to high acoustic variability and poor transcription accuracy. Below, we outline the regional dialects for our three target languages, identify the specific variants chosen for the **Kingdom Communication Solutions (KCS) Media Studio**, and explain the reasoning behind these choices.

2.1.1 Ewondo

Ewondo, part of the Beti-Pahuin group (Guthrie A70) in the Centre Region, features several regional variations [32]. The dominant urban dialect is Ewondo Yaoundé, spoken primarily in the capital and characterized by French lexical borrowings [30]. Further south, along the Nyong river, the Mbalmayo variant exhibits distinct vowel lowering [18]. In the northern periphery, the Mvele and Yezum dialects display unique tonal contours [33], while the eastern forest margins are home to the Bane and Yabekolo transitional dialects.

For this project, we selected **Ewondo Yaoundé** as our standard variant. It is widely accepted across regional media and education, making it the most stable choice. Additionally, it possesses the largest volume of documented text, which helps reduce vocabulary errors when training acoustic models.

2.1.2 Ghɔmálá'

Ghɔmálá' is a major Eastern Grassfields language spoken in the West Region. Its central variety is the Bandjoun (Jo) dialect, which holds historical prestige [28]. The Bafoussam (Fusap) variant serves as the commercial hub dialect, known for rapid speech and word elision [31]. The western and southern areas feature the Baham (Hom) and Batié (Te') varieties, with distinct vowel shifts [34], while highland dialects like Bamougoum (Mûgum) and Bansoa (Ssa') retain older consonant structures [17].

We have chosen the **Bandjoun** dialect for the platform. As the foundational standard for written Ghɔmálá', it is well-supported by existing dictionaries and literacy materials, ensuring that our transcriptions align with established written norms.

2.1.3 Basaa

Basaa is a Narrow Bantu language (Guthrie A40) spoken across the Littoral and Centre Regions. Its main dialect is Basaa Central (Nyangon), which connects the Sanaga-Maritime and Nyong-et-Kéllé divisions [19]. Other variations include the Bakem and Bon transitional dialects in the north and west [20], the Bassa de Babimbi forest cluster along the Sanaga river [35], and the southern coastal dialects (Log and Mpo) that show noun-class simplification [36].

We selected **Basaa Central** because it offers the highest lexical consistency with major reference lexicons, such as the *Dico Bassa'a*. Its balanced phonology also minimizes transcription conflicts when collecting audio from diverse speakers.

2.2 Data Collection Methodology

Building a reliable corpus for the KCS Media Studio requires more than just gathering audio; it demands rigorous validation to prevent speech recognition failures caused by literal or unverified translations. To achieve this, we use a hybrid data collection model combining active community fieldwork with structured remote verification [24].

Our collection workflow is organized into two parallel phases:

Textual Translation and Verification: Initial drafts are translated by fluent native speakers instructed to prioritize natural contextual meaning and idiomatic phrasing over word-for-word literal translations. Once drafted, these texts are systematically cross-referenced against established lexical databases,

Table 3: Methodological Mapping Matrix: Research Questions, Problem Statements, Nunamaker Phases, Objectives, and Evaluation Metrics

RQ	PS	Nunamaker Phase	Objective	Target Artifact / Deliverable	Evaluation Metric / Criteria
RQ1	PS2	Observation	**RO1.1.O**	Sociolinguistic Dialectal Audit Report	Translator pain-point taxonomy
		Theory	**RO1.2.T**	Hierarchical Dialect Mapping Taxonomy	Semantic coverage of regional lexicons
		Implementation	**RO1.3.I**	Adaptive Dialect Toggle and Suggestion UI	Feature completeness in KCS Studio
		Evaluation	**RO1.4.E**	Comparative Translator User Study	NASA-TLX score, Dialect acceptance rate
RQ2	PS1	Observation	**RO2.1.O**	Browser Waveform and Latency Benchmark	Synchronization offset analysis (ms)
		Theory	**RO2.2.T**	9-Parameter Audio Metadata Protocol	Schema validation rate, Zero collisions
		Implementation	**RO2.3.I**	Full-Stack KCS Media Studio Web Platform	End-to-end video captioning pipeline
		Evaluation	**RO2.4.E**	Controlled Transcription Benchmark Study	WER, CER, Throughput time (min/video)

standardized GACL orthographies, and published dictionaries.

Field Audio Capture: Native audio recordings are gathered directly within the respective linguistic regions. Fieldworkers deploy custom KoboToolbox digital forms on Android mobile devices [37]. The form sequentially guides the interaction: logging speaker demographic metadata (age tier, biological sex, dialect variant) and consent, displaying the target concept from our standardized 363-word lexicon, and recording uncompressed audio (16-bit 44.1 kHz PCM) directly via the device microphone. Because KoboToolbox operates fully offline, records are securely stored locally and synced in batches to our central cloud repository upon establishing internet connectivity.

2.3 Consent and Validation Procedure

Ethical compliance and rigorous quality control form the foundation of our corpus pipeline. This ensures that every linguistic contribution is authorized, ethically gathered, and linguistically sound before entering the system repository.

Informed Consent Protocol. Prior to any recording, field teams secure documented informed consent from all participating native speakers. Participants are fully briefed on the project scope, digital archiving of their speech data, and intended use within the KCS Media Studio platform. Contributors retain full autonomy, including the explicit right to withdraw their data at any stage of the research process without penalty [23].

Independent Double-Validation Framework. To ensure high semantic fidelity and orthographic consistency across Ghɔmálá', Ewondo, and Basaa, raw submissions bypass direct repository ingestion [6]. Instead, every submission enters a mandatory, multi-tiered validation loop:

1. **Primary Review**: Conducts an initial technical audit, checking for basic audio clarity, signal-to-noise ratio, and timestamp integrity.
2. **Independent Secondary Audit**: Performed by an independent second native speaker who evaluates the submission for contextual accuracy, idiomatic phrasing, and phonetic correctness against reference dictionaries.
3. **Dispute Resolution**: In instances of dialectal disagreement, a senior linguistic supervisor adjudicates

the correct orthographic representation.

4. **Commitment Stage**: Ensures the dataset is committed to the project pipeline only after receiving formal dual validation.

2.4 Audio File Naming Convention and Corpus Architecture

To ensure systematic indexing, strict traceability, and seamless ingestion into automated speech recognition pipelines, raw audio recordings gathered during fieldwork adhere to a standardized nine-parameter naming convention:

`ISO-Lang_Variant_Demographics_Content_-Environment_Location_Timestamp_ConceptID_-UID.wav`

Structural Breakdown of the Naming Parameters

ISO-Lang (ISO 639-3 Language Identifier): A standardized three-letter code designating the target language (`ewo` for Ewondo, `bbj` for Ghɔmálá', `bas` for Basaa).

Variant (Regional Dialect Code): A concise descriptor identifying the specific dialectal standard (`yao` for Ewondo Yaoundé, `bdj` for Bandjoun, `ctr` for Basaa Central).

Demographics (Age Group and Sex): An anonymized demographic token capturing the speaker's categorical age bracket and biological sex. Sex is recorded as `M` (Male) or `F` (Female), paired with standardized generational age tiers (`YTH` for youth aged 18–29, `ADL` for adults aged 30–54, and `SRV` for seniors aged 55 and above). For example, `ADL-F` represents an adult female.

Content (Lexical or Discourse Type): A categorical tag classifying the utterance type (`LEX` for structured 363-word items; `NAR` for narrative passages).

Environment (Acoustic Condition): An environmental descriptor aiding preprocessing noise-filtering algorithms (`STU` for studio; `QUI` for a quiet indoor room; `FLD` for ambient field settings).

Location (Recording Origin): A three-letter geographic airport/city code indicating where recording occurred (e.g., `YAO` for Yaoundé, `DLA` for Douala, `BFN` for Bafoussam).

Timestamp (Acquisition Date): Exact capture date formatted strictly as `YYYYMMDD` for chronological versioning.

ConceptID (Lexicon Index): Sequential concept index mapped to our predefined 363-word lexicon (`ASD-0001` through `ASD-0363`).

UID (Unique Identifier): A cryptographically generated alphanumeric string (e.g., short UUID or hash) preventing accidental overwrites.

Practical Application Example

An audio recording captured on September 11, 2026, in Yaoundé, where an adult male speaker articulates the first word from our standardized lexicon under quiet indoor conditions, assigned UID `1a2b3c4d`, yields:

`ewo_yao_ADL-M_LEX_QUI_YAO_20260911_-ASD-0001_1a2b3c4d.wav`

3 Expected Artefacts

This project generates three primary categories of deliverables:

3.1 The Technological Artefact

The core software deliverable is the **Kingdom Communication Solutions (KCS) Media Studio**. Engineered as a cloud-native web platform (React frontend, Web Audio API, Canvas waveform scrubbing, and FastAPI backend), it provides AI-assisted transcription, split-track dialect toggling, and synchronized WebVTT caption editing specifically adapted to GACL character encodings and tonal diacritics.

3.2 The Linguistic Data Artefact

A foundational outcome is the creation of standardized, double-validated digital speech corpora for Ghɔmálá', Ewondo, and Basaa. Comprising phonological dictionaries, time-aligned native-speaker audio pairs, and metadata manifests, these corpora provide independent, open resources for training acoustic and text-to-speech models.

3.3 The Methodological Artefact

The final deliverable is the formal data collection and validation protocol itself. Integrating offline KoboToolbox mobile workflows, documented informed consent, and multi-tiered linguistic validation, this framework provides a replicable blueprint for future indigenous language preservation initiatives across Africa.

4 Conclusion

The severe deficit of digital speech datasets and localization tools for indigenous Cameroonian languages represents a critical barrier to computational equity. Through the engineering of the Kingdom Communication Solutions (KCS) Media Studio, this project establishes a foundational technical and methodological framework integrating Ghɔmálá', Ewondo, and Basaa into modern AI multimedia pipelines. By uniting field-validated acoustic data collection protocols, strict sociolinguistic metadata structuring, and an adaptive browser-based editing environment, this research provides an essential stepping stone toward equitable speech recognition and language technology for African cultural heritage.

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